Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. TEAM AND PROJECT OVERVIEW

Team Name

Calador

Selected Domain

Domain 1 - The Aiversecity Future-Ready Workforce Domain

Selected SDG Track (1–17)

SDG 4 Quality Education (main); also SDG 5, 8, 9, 10

Project / Solution Name

Calador - AI Career & Skills Navigator

Name (Leader)

Soon Yen Ling

Email & Phone (Leader)

soonyling5@gmail.com & 0177281683

University Affiliation(s)

TARUMT, UTAR

Number of Team Members

4

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysia's workforce is adopting AI faster than it is being trained for it. The Domain 1 brief cites the Hays Asia Salary Guide 2025: 56% of Malaysian professionals already use AI tools at work, yet only 26% have received formal training. The Ministry of Science, Technology and Innovation (MOSTI) has said more than 600,000 workers will need reskilling within three to five years.

Those affected are working adults at risk of job displacement, fresh graduates and career switchers choosing where to go next, HR and learning-and-development managers deciding which roles to train their teams for, and TVET and university career centres advising students. They face three gaps: workers do not know which AI skills are worth learning, employers cannot tell which roles to train for, and learning pathways are static lists that are out of date when published.

This matters economically, because under-prepared workers and graduates limit productivity and decent employment (SDG 8). It matters socially, because people with the least access to guidance are most exposed to displacement (SDG 10), and relevant skills training is the core of SDG 4. It also matters for sustainability: many future roles in responsible AI, health, clean energy and agriculture will help deliver the 17 SDGs. Because AI progress is unpredictable, a one-time skills assessment is not enough; people need guidance that keeps updating.

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Calador is a Streamlit web app that takes a user from "who I am today" to "where I can be in the AI era" in minutes. It is not a chatbot: no large language model is used. Every output is computed from rules and public data, so nothing is invented and each result cites its source.

1. Skills profile. Form, CV upload (PDF/DOCX/TXT, processed in memory only) and a short potential questionnaire (weekly hours, learning style, willingness to learn, SDGs of interest).

2. Future Skills Radar. Compares 2025 vs 2026 skill and role shares in the Kaggle AI Jobs Market dataset, then adds the latest arXiv cs.AI papers and live Arbeitnow job ads into a skill-momentum score, each with source and fetch date.

3. Role matching. Match score = 90% skill coverage + 10% background relevance, with a skills-gap breakdown. Trend signals only fine-tune ranking, and the app shows when recommendations re-plan because trends changed.

4. Reskilling pathway. Ordered by skill momentum, using free resources (Kaggle Learn, Google MLCC, Elements of AI, HRD Corp e-LATiH), with estimated hours, progress tracking and badges.

5. SDG mapping. Every role and skill is tagged to SDGs, with a chain showing how the chosen path contributes.

6. Uncertainty. Slow, moderate and rapid-disruption scenarios with adjustable ranges, labelled as estimates.

Also built: HR / career-centre batch mode with a team skills-gap heatmap, a fairness check, and an English / Bahasa Malaysia / Chinese interface with a low-bandwidth mode. Workshop skills used: cloud deployment of the app for Phase 2 and data handling with open datasets and public APIs.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

App and UI: Python, Streamlit, Altair charts, custom CSS; mobile-friendly, with a low-bandwidth mode (tables instead of charts, no live fetch).

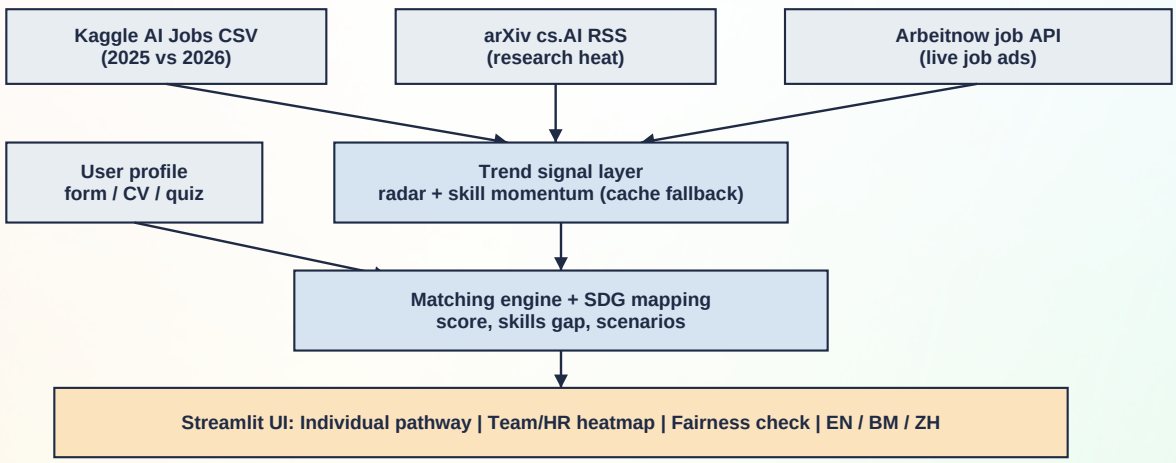
Logic: pandas and NumPy; rule-based role matching, SDG mapping and skill-momentum scoring. No LLM.

Updatable trend sources: Kaggle AI Jobs Market 2025-2026 CSV (2025 vs 2026 shares), arXiv cs.AI RSS via feedparser, Arbeitnow public job-board API. Each source shows its URL and fetch date; local JSON caches are the fallback if a feed fails.

CV intake: pypdf and python-docx; text is read in memory and not stored.

Official sponsor GenAI API or cloud platform: not used in this prototype (no LLM by design, to prevent invented jobs or courses).

System block diagram



3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise: we started from the four user groups in the Domain 1 brief (working adults, graduates and switchers, HR/L&D managers, career centres) and the gap they share: nobody knows which skills to learn next.

Define: one clear journey, profile to radar to ranked roles to pathway, with every recommendation tied to a visible source.

Ideate: rule-based, explainable scoring instead of a chatbot, so the system cannot invent jobs or courses.

Prototype: a working Streamlit app built on the Kaggle dataset, arXiv and Arbeitnow feeds, with cached fallbacks if a feed is down. Only public or synthetic data is used; CVs are processed in memory.

Test: a built-in fairness check compares scores across backgrounds with identical skills. In Phase 2 we will run usability tests with students and an HR/career-centre user, then fix the friction found.

Deploy: public cloud URL and GitHub repository, with low-bandwidth mode and three languages for access on mobile.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Target 4.4:** substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship. Supporting: **4.3** (lifelong technical and vocational learning), **8.2** (productivity through innovation) and **8.5** (full and productive employment).
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	**26%** of Malaysian professionals have received formal AI training, while **56%** already use AI tools at work: a **30-percentage-point training gap**. More than **600,000** workers need reskilling within three to five years (MOSTI). Sources: Hays Asia Salary Guide 2025 and MOSTI statement, as cited in the Domain 1 brief (Background section).

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	**Pathway coverage:** % of a user's identified skill gaps matched to a cited free or low-cost learning resource. Skills with no match are shown as "no resource", never invented. Secondary KPIs: minutes from profile entry to ranked roles and pathway; number of trend sources shown with source and fetch date.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	By the final prototype: **at least 80%** of gap skills across the dataset's roles matched to a cited free or low-cost resource; profile to pathway in **under 5 minutes** in user tests; all 3 trend sources shown with source and date. Starting coverage will be measured on the dataset's skill vocabulary at the start of Phase 2.

5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Kaggle: AI Jobs Market 2025-2026 (alitaqishah)	Dataset (job postings CSV)	As stated on the Kaggle dataset page
arXiv cs.AI RSS feed; Arbeitnow job-board API	Public data feeds (research paper titles; live job ads), fetched at run time and cached locally	Subject to arXiv and Arbeitnow terms of use; used as public data only
Streamlit; Altair	Code (open-source frameworks)	Apache-2.0; BSD-3-Clause
pandas; NumPy; feedparser	Code (open-source libraries: data handling, RSS parsing)	BSD-3-Clause; BSD-3-Clause; BSD-2-Clause
pypdf; python-docx	Code (open-source libraries: CV text extraction)	BSD-3-Clause; MIT

6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature] *Soon*

[Date] 24 October 2026